



RICE FARMERS' UTILIZATION OF INFORMATION COMMUNICATION TECHNOLOGIES IN WESTERN AGRICULTURAL ZONE OF BAUCHI STATE, NIGERIA

*Itse, V., 1Abdullahi, S. and 2Abu, A. I.

1 Department of Agricultural Extension and Rural Development, Abubakar Tafawa Balewa University Bauchi, Bauchi State, Nigeria.

2 Department of Agricultural Extension Services, University of Maiduguri, Borno State, Nigeria.

*Corresponding Author: victor.itse70@gmail.com, (07037711270)

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ABSTRACT

Key words:

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The study examined the utilization of information and communication technologies among small-scale rice farmers in Western Agricultural Zone of Bauchi State, Nigeria. Multi-stage random sampling techniques were used in selecting 250 respondents. Data were collected using structured questionnaire and analyzed using descriptive and inferential statistics. The result showed that the mean age of the farmers was 43 years. Majority (67% and 84.4%) of the farmers were male and married. The mean household-size was 9 persons and majority (85.4%) had formal education. The mean farming experience of 15 years with average monthly income of N41,274.03k. Radio ($\bar{x}=4.13$) and mobile phone ($\bar{x}=3.69$) were the most frequently used ICT facilities in accessing rice farming information by the farmers. The result of logistic regression analysis showed that level of education and extension contact were positive and significant at ($P<0.01$) while frequency of extension contact was negative and significant ($P<0.05$) in influencing the utilization of ICTs. The major constraints to ICT usage were erratic power supply (95.8%), inadequate training and practical exposure to some ICTs (94.1%) and high cost of modern ICTs (93.4%). The study concluded that level of education and extension contact were the factors influencing rice farmers utilization of ICTs. The study recommended that agricultural extension agencies should sensitize farmers on uses of mobile phone and radio in carrying out ICT-based extension-services, government agencies and other stakeholders should provide adequate ICT supporting facilities to encourage its widespread utilization.

INTRODUCTION

Information is vital for rural and agricultural development. Information and communication technologies (ICTs) are new technologies that cannot be ignored in Africa especially for development in all sectors, agriculture inclusive. This is because, ICTs are the main driving forces than can bring about development and change in this digital age (Olaniyi, 2014). The recent development in ICTs has broken international and national barriers and turned the world into a global village, making information available to everyone, everywhere and at any time it is needed (Onasanya *et al.*, 2011). Information and Communication Technologies (ICTs) are the devices that help to connect people via different medium to exchange information among people (Nwafor *et al.*, 2020). The ICT devices that help facilitating farming activities encompassed applications like radio, television, cellular phones, computers, tablets and networking, hardware and software, satellite systems (Pande and Deshmukh, 2015). ICTs are required to sensitize, mobilize, persuade, and

sustain farmer's interest with a view to adopting new ideas, practices or technologies that are likely to bring about holistic improvement in the well-being of individuals, thereby enabling them to escape from the trap of poverty.

In Nigeria, agriculture is the principal industry and source of income for more than 70% of the people (Onwutuebe, 2019). Peasants make up more than 85% of the population, and small-scale agriculture is the mainstay of their rural economy. Rural Nigerians' primary source of income is still small-scale agriculture, which is characterized by low capital investment, low productivity, and a focus on subsistence. Growing urban populations and the reduction of arable land, increase the need for productive, efficient, and environmentally friendly ways of agricultural production. Today, farmers are plagued by myriads of problems that are becoming specialized and complex, farmers are often unfamiliar with the modern agricultural practices or new technologies (Norsida *et al.*, 2016). These challenges restrict their income and could hindered adoption of improved technologies with attendant consequences of poor performance of the agricultural sector. Information and communication technologies (ICTs) play an important role in addressing these challenges and improve the livelihoods of the rural poor people. Rehman *et al.*, (2016) opined that these digital technologies would enhance food production and ensure sustainability in the agricultural sector. ICTs utilization and precision agriculture tools have resulted in increased efficiency and reduced costs. It has also delivered decision-making tools that boost agricultural productivity as well as help manage natural resources effectively.

However, aside from the existence of these technologies, there has been the problem of accessibility for farmers and many others who may want to utilize them for agricultural purposes (Mtega and Msungu, 2013). Several studies (Richardson 2006, Yekini & Hussein 2007, Balboni, 2010 and Dunne (2016) have pointed out the need for adoption of ICTs for enhancing development. It against this backdrop the research wishes to be conducted. In view of the above, this study seeks to provide answers to the following research questions; what are the socio-economic characteristics of the respondents in the study area? what is the level of ICT usage by the respondents? what are the factors influencing the use of ICT facilities in the area? and lastly, what are the constraints to utilization of ICTs among respondents in the area?

Objectives of the Study

The broad objective of the study was to examine the utilization of Information and Communication Technologies (ICTs) among small-scale rice farmers in boosting agricultural production in Western Agricultural Zone of Bauchi State, Nigeria. The specific objectives were to:

- i. describe the socio-economic characteristics of the respondents in the study area,
- ii. determine the level of ICTs usage in the study area,
- iii. determine the factors influencing ICTs usage and
- iv. describe the constraints to utilization of ICTs by the farmers in the study area.

METHODOLOGY

i. The Study Area

The study area is Western Agricultural Zone of Bauchi State, Nigeria. Bauchi state comprises of three Agricultural Zones namely: Northern Northern Zone (9 LGAs) Central Zone (4 LGAs) and Western Zone (7 LGAs). The 7 Local Government Areas in Western Zone are namely; Bauchi, Kirfi, Alkalari, Toro, Dass, Tafawa Balewa and Bogoro LGAs with Bauchi as the State capital. The study area is located between latitudes 9°30' and 10°48' North of the equator, and between longitudes 8°45' and 15'. East of the green-which meridian. The projected population in 2006 based on the 3.6% growth rate is estimated at 2,591,767 million people representing (31.5%) of the entire States' population (NPC, 2018 Projection). The area has a total land mass of 49,259 sq kilometers. Bauchi State is bordered in a clockwise direction by Yobe, Gombe, Taraba, Plateau, Kaduna, and Kano and Jigawa State respectively. The major occupations of the clientles include farming, livestock rearing and trading.

ii. Sampling Technique and Sample Size

A multi-stage random sampling techniques was used in selecting the respondents. First stage was purposive selection of western agricultural zone Bauchi State because of the intensity of rice production. Second stage was random selection of five (5) Sub-zonal offices from each Local Government Areas in the Zone. Third stage was random selection of two (2) Blocks Extension Offices based on their intensity of rice production from the Sub-zonals, making a total of ten (10) blocks. Fourth stage was purposive selection of two (2) Cells from each of the ten (10) Blocks Extension Offices making a total of 20 Cells. Finally, in the fifth stage, 5% of the total sampling frames in each of the five sub-zonal areas were purposively selected to make a total of 250 respondents. In determining the sample size for this study, a model was used, where 20% was suggested when a population size is up to 1,000. 10% was suggested when the population size is up to 5,000 and 5% when the population size is up to 10,000 (Alamu and Olukosi, 2010). According to this model, the appropriate sample frame is 5,002 registered rice farmers and sample size used for this study was 250 which constitute 5% of the randomly selected registered rice farmers as illustrated in Table 1 below.

Table 1: Sample size selection plan for the study

Sub-zonal Areas	Blocks	Cells	Sample Frame	Sample Size (5%)
Bauchi	Bauchi	Bauchi	550	27
		Durum	435	22
	Kangare	Kangare	435	22
		Dindima	318	16
Sub-total			1,738	87
Alkalari	Alkalari	Alkalari	170	8
		Masuri	175	9
	Duguri	Duguri	185	9
		Gaji	152	8
Sub-total			682	34

Dass	Dass	Dott	105	5
		Dabardak	93	5
	Bazali	Badel	90	5
		Wandi	102	5
Sub-total			390	20
Toro	Tilde	T/Fulani	190	9
		Dirikko	182	9
	Zaranda	Nabordo	230	12
		T/Giwa	255	13
Sub-total			857	43
Tafawa Balewa	T/ Balewa	T/ Balewa	385	20
		Shall	320	15
	Bununu	Bununu	330	18
		Burga	300	14
Sub-total			1,335	67
Total	10	20	5,002	250

Source: Field survey (2022)

iii. Data Collection

Data were collected from primary and secondary sources. The primary data was collected by using structured questionnaires to obtain information from the respondents. Based on the specified objectives, the questionnaire was categorized into sections. Journals, the internet and government publications were some examples of secondary sources of knowledge. Two hundred and fifty (250) questionnaires were administered at random to the respondents. However, out of the questionnaires distributed 212 were completed retrieved. This represented about 85.0% of the questionnaire and hence, the data were collected with these 212 obtained.

iv. Data Analysis

Data were analyzed using descriptive and inferential statistics. The descriptive statistics used were means, frequency and percentages to analyze objectives (i),(ii) and (iv) while multiple regression analysis was inferential statistics technique used to analyze objective (iii).

The regression model was implicitly expressed as follows:

$$Y = f (X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8, X_9, X_{10}, e) \text{ -----}$$

(1)

Explicitly, the model is specified as;

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4 + b_5X_5 + b_6X_6 + b_7X_7 + U \dots\dots\dots$$

(2)

Where:

Y = (Utilization of ICT which is a dummy variable 1 for utilization and 0 for otherwise)

a = Constant

b₁ - b_n = Regression Coefficients

X_1 = Age of respondents (Years)
 X_2 = Sex (1 if male, 0 if female)
 X_3 = Marital status (Married = 1; Single = 0)
 X_4 = Household size (number of persons)
 X_5 = Level of Education (Years of formal education)
 X_6 = Farming Experience (Measured in Years)
 X_7 = Farm size (Measured in Hectares)
 X_8 = Annual Income (Naira)
 X_9 = Contact with Extension Agent (dummy variables, 1=yes, 0=No)
 X_{10} = Frequency of contact (Number of times visited by Ext. Agent)
 X_{11} = Membership of Association (dummy variables, 1=yes, 0=No)
 X_{12} = Access to Credit (dummy variables, 1=yes, 0=No)
 U = Stochastic error term.

RESULTS AND DISCUSSION

i. Socioeconomic Characteristics of Respondents

The result in table 2 reveals that majority (71.4%) of the respondents were within the age group of 28-57 years with a mean of 43 years. The fact that the small-scale rice farmers were generally in their middle years and in a stage of active, productive, and economic viability suggests that this could have motivated them to utilize ICTs. This finding corroborates with Nuer (2018) who reports that ICT is considered to entice the youth to engage in agribusiness and make agriculture more attractive to them. The result also indicates that majority (67.0%) of the respondents were male, while one-third (33.0%) were female. This implies that men were found to be more active participants than their female counterpart in the use of ICTs for farming in the study area. This concurs with the result of Machina and Lubungu (2018) who reported that male-headed households have higher access to productive resources and information that increases their chances of using ICTs in farming.

The result further showed that majority (84.4%) of the respondents were married, while 15.6% were single. This implies that more married persons are more disposed towards using ICTs in small-scale rice production in the study area. This may be attributed to their need to stay in touch and communicate more. This is in tandem with the findings of Nwokoro and Chima, (2017) that larger percentage of married persons could be attributed to the higher patriarchal orientation in rural settings. Which tends to give married individuals an edge over unmarried ones such that unmarried men and women are denied some entitlements to rural assets like land and livestock which constitute essentials of their livelihood. The mean household size was 9 persons. This shows that respondents had household above the national average of 5 persons according to (Kassie 2017). The implication of majority (71.7%) of the respondents having 1-10 persons in their households is that the varied ICTs skills could be employed. This is in line with the findings (Njuguna *et al.*, 2015) that families with higher household sizes have the likelihood of adopting improved farming practices since they are believed to have adequate labour for farming activities and can share new farming skills which any member of the household has acquired.

The results on educational status shows that majority (70.8%) of the respondents had formal education. This implies most respondents were literate and can therefore grasp and adopt productive information disseminated via ICTs for various agricultural and livelihood

activities. This finding is in agreement with position put forward by Zhuang and Naminse (2018) that education and technical training are important variables that could enhance farmers' use of ICT to boost their production and allow people to adapt more easily to both social and technical changes. However, the result (70.8%) on education aligns with the findings of Gautam and Andersen (2017), and Igwe (2018) who observed similar results.

Furthermore, the average years of farming experience was 15 years. This implies that respondents have gained reasonable number of years in rice farming that and can be said to have acquired the wherewithal to support production that can allow for effective utilization of the benefits of new technologies to improve their socioeconomic lives in the study area. This assertion is amplified by Onemolease *et al.*, (2021), who opined that such experience impacts farmers' managerial know-how and productivity. The result on age of experience agrees with the finding of Titilope (2017) who reported similar mean. The result further reveals a mean farm size of 1.5 hectares, implying that majority of the respondents are small-scale rice farmers. This corroborated with the report of Thomas *et al.* (2017) that farm size plays a critical role in adoption process of a new technology. Furthermore, the result showed that majority (65.6%) of the respondents had access to extension contact. This implies that the respondents were exposed to technical innovations that could enhance their farming practices for increase yield and income in the stud area. The result agrees with the report of Odoemenem and Obine (2015) that through extension visits, farmers are better informed about new technologies which could increase their productivity.

Also the result on frequency of extension visit revealed that majority (61.3%) had monthly visitation by the extension workers. This suggests that a significant portion of the respondents use these interactions to increase their agricultural output in the study area. This is also in line with the findings of Odoemenem and Obine (2015), who noted that frequent meetings and extension contact between extension staff and farmers would inform and raise farmers' understanding of the possible benefits of enhanced agricultural technologies. Also, the result revealed that majority (91.0%) of the respondents earned average of ₦41,274.03 monthly. This shows that the respondent's annual income was relatively low and that a household head with limited resources could prefer to spend money on necessities like food, healthcare costs, and school expenses rather than buying and maintaining ICT equipment like phones, computer hardware, and televisions. When their advantages are fully realized, ICTs are thought to increase farmers' revenue. This result is consistent with the findings of Chikaire *et al.*, (2017), who noted that there is a crucial relationship between farmers' income levels and ICT use because, whenever income levels are likely to rise, information access is likely to be simple, which may facilitate the making of wise agricultural decisions.

Table 2: Distribution of the Respondents based on Socioeconomic Characteristics (n=212)

Variable	Frequency	Percentage	Mean($\bar{x}$)
Age(Years)			
18-27	25	11.8	43
28-47	109	51.4	
48-57	42	19.8	
58-67	36	17.0	
Sex			
Male	142	67.0	
Female	70	33.0	

Marital Status			
Married	179	84.4	
Single	33	15.6	
Household Size (No. of Persons)			
1-10	134	63.2	9
11-15	39	18.4	
16-20	32	15.1	
21-25	7	3.3	
Level of Education			
Never attended	5	2.4	
Adult education	31	14.6	
Qur'anic School	26	12.3	
Primary School	22	10.4	
Secondary School	68	32.1	
Tertiary Institution	60	28.3	
Farm Size(ha)			
0.5-1.5	118	55.7	1.5
1.6-2.5	60	28.3	
2.6-3.0	34	16.0	
Farming Experience (Years)			
1-10	83	39.2	15
11-20	80	37.7	
21-30	32	15.1	
31-40	17	8.0	
Extension Service			
Yes	139	65.6	
No	73	34.4	
Frequency of Contact			
None	74	34.9	
Weekly	30	14.2	
Fortnightly	31	14.6	
Monthly	69	32.5	
Annually	8	3.8	
Income Level (₦)			
₦ 10,000 - ₦ 20,000	24	11.3	41,274.03
₦ 20,001 - ₦ 30,000	38	17.9	
₦ 30,001 - ₦ 40,000	22	10.4	
₦ 40,001 - ₦ 50,000	37	17.5	
> ₦ 50,001	91	42.9	

Source: Field Survey (2022)

ii. Extent of Usage of ICT Tools

Table 3, shows the distribution of the respondents based on the usage of ICT tools. Findings on frequency of utilization of ICTs is shown. The result shows that respondents utilized radio ($\bar{x}$ = 4.13) and mobile phone ($\bar{x}$ = 3.69) to a high extent in their farming activities, since their mean scores were greater than the cut-off point of ($\bar{x}$ = 3.00). Others were not frequently utilized in the study area as their means score was below the decision point. This implies that the conventional ICT tools such as radio and mobile phone were the most frequently used ICTs for obtaining agricultural information by the respondents in the study

area than the contemporary ones such as computer and internet. The possible reason could be because of their ease of use, wide coverage and alternative source of power (battery) systems. This finding is in tandem with the report of Lokeswari (2016) that only radio and mobile phones were found to be the most ICTs used in receiving agricultural information.

Table 3: Distribution of the respondents based on extent of usage of ICT tools.

ICT Tools	Extent of Usage			Mean*	Remark
	HU	LU	NU		
Radio	172(81.1)	28(13.2)	12(5.7)	4.13	High
Mobile Phone	142(67.0)	39(18.4)	31(14.6)	3.69	High
TV/Satellite	40(18.9)	72(33.9)	100(47.2)	2.19	Low
Internet	29(13.7)	108(50.9)	75(35.4)	2.11	Low
VCD/DVD	3(1.4)	85(40.1)	124(58.5)	1.48	Very Low
Computer	13(5.42)	199(93.9)	1(0.68)	1.43	Very Low
Print media	11(5.2)	39(18.4)	162(76.4)	1.41	Very Low

Source: Field Survey, 2022

Note: Figures in parenthesis are percentages

Key: High utilization = **HU**, Low utilization = **LU**, No utilization = **NU**

*Any mean greater than ≥ 3.0 threshold is significant and mean < 3.0 threshold is not significant

iii. Factors Influencing Respondents' Utilization of ICT Tools

The result of logistic analysis of regression analysis showed that the pseudo R^2 of 0.23 which implies that 23% of the variation in the dependent variable was explained by the independent variables included in the model. The result is presented in Table 4. The coefficient of education and extension contact were positively significant at 1% level of probability implying that the higher the educational level and extension contact, the higher the chance of utilization of ICTs. The coefficient of frequency of extension contact was negatively significant at 5% meaning that frequency of extension contact decrease in farmer's likelihood to use ICTs. This implies that the number of years spent in school and constant exposure of farmers to extension trainings will increases awareness and adoption of ICTs among respondents. This is in line with *a priori* expectations. This finding agrees with the report of Rehman *et al.*, (2016) that different factors which may be social, economic, technical or cultural influence the use of different information sources by the farming community.

Table 4: Distribution of Respondents by Factors Influencing ICTs Utilization

Variables	Regression Coefficient	Standard error	t-values	Result
(Constant)	14.850	1.562	9.507	.000
Age	-.009	.029	-.315	.753NS
Sex	.506	.508	.996	.320NS
Marital Status	-.714	.706	-1.012	.313
Household Size	.010	.052	.186	.852 NS
Level of Education	.456	.157	2.898	.004***
Major Occupation	-.022	.032	-.668	.505
Farm Size	.107	.373	.027	.774
Income Level	.045	.205	.222	.824

Extension Contact	3.468	.898	3.864	.000***
Freq. of Ext. Contact	-.623	.295	-2.110	.036**
Membership of Association	.457	.500	.915	.361
Credit access	.470	.711	.059	.509
Pseudo R²	0.23			

Source: Field Survey, 2022

Note: ***=Significant at 1% level of probability, **=Significant at 5%, *=Significant at 10%, NS = Not Significant

iv. Constraints of the respondents to the Usage of ICT Tools

The result as shown in Table 5 identified constraints to the use of ICTs by small-scale rice farmers in the study area. Low network connectivity ($\bar{x}$ = 4.28), poor/erratic power supply ($\bar{x}$ = 4.13) and inadequate training and practical exposure towards ICTs ($\bar{x}$ = 4.10) were the foremost constraints militating against the use of ICTs and are ranked (1st, 2nd and 3rd), while Poor finances (x = 3.93) and language barriers (x = 3.90), which were ranked 12th and 13th, respectively, were the least significant factors that the respondents believed had prevented them from using ICT facilities in the study area effectively. This result is in tandem with the report of Adamu *et al.*, (2017) who identified poor power supply, inadequate ICT Facilities, high cost of ICTs tools, insufficient infrastructure, poor network connectivity are challenges that obstruct successful implementation and use of ICT.

Table 5: Distribution of the Respondents' based on Constraints to the Usage of ICTs in rice farming (n=212)

Perceived Constraints	A	UD	DA	($\bar{x}$)	Rank
Low network connectivity	194(91.5)	15(7.1)	3(1.4)	4.28	1 st
No light/erratic power supply	193(91.0)	16(7.5)	3(1.4)	4.13	2 nd
Lack of training & practical exposure to ICT	191(90.1)	17(8.0)	4(1.9)	4.10	3 rd
High cost of ICT tools	198(93.4)	7(3.3)	7(3.3)	4.08	4 th
High cost of repairing ICTs	194(91.5)	13(6.1)	5(2.3)	4.07	5 th
Lack of repairing facilities & centers in villages	184(86.8)	18(8.5)	10(4.7)	4.04	6 th
Theft and insecurity	186(87.8)	11(5.2)	15(7.0)	4.00	7 th
Negative attitude towards ICTs	174(82.1)	24(11.3)	14(6.6)	3.98	8 th
Inadequate awareness of benefits of ICTs	184(86.8)	18(8.5)	10(4.7)	3.96	9 th
Low ICT literacy	182(85.8)	16(7.5)	14(6.6)	3.95	10 th
Lack of skill in handling ICTs	174(82.1)	14(6.6)	24(11.3)	3.94	11 th
Poor Finance	174(82.1)	24(11.3)	14(6.6)	3.93	12 th
Language Barrier	177(83.5)	26(12.3)	9(4.3)	3.90	13 th

Source: Field Survey (2022)

Figures in Parenthesis are percentages.

Key: A= Agreed, UD = Undecided, DA= Disagree

* Any mean score < 3.00 threshold is not significant, while ≥ 3.00 threshold is significant or termed true

CONCLUSION

It is possible to infer from the results that the level of education and extension contact were factors impacting ICT use among small-scale rice farmers. The importance of raising educational standards in the study area cannot be overstated because it has the potential to increase respondents' use of ICT and, consequently, offer a solution to the shortage of extension agents.

RECOMMENDATIONS

Based on the findings of the study, the following recommendations were made:

- i. Agricultural extension agencies in Bauchi State should sensitize farmers on uses of mobile phone and radio in carrying out ICT – based extension services.
- ii. Government in general and other stakeholders need to provide adequate network coverage and the necessary ICT supporting facilities to encourage its sufficient widespread and utilization.
- iii. Governments and concern non-governmental agencies should put in place a digital policy at all levels to ensure that staff of public agricultural institutions possess the requisite skills to use ICTs.
- iv. Local governments should construct ICT training centers for farmers so that these people can learn how to use ICTs.
- v. The government should make ICT devices accessible and subsidize them at the local level so that small-scale farmers can access information to better their livelihoods.

REFERENCES

- Adamu, S., Jibo, I. Mohammed, A. & Saidu, A. (2017). Application of ICT in Agriculture, *International Journal of Computer Science and Mathematical Theory*, 8(1):8-18.
- Alamu, J. F. and Olukosi, J. O., (2010). Simplified Research Methodology: Principles and Practices (Rev. ed). Zaria Great Glory Publishers.
- Balboni M., (2010) Progress and challenges in measuring ICT in Latin America and the Caribbean, International Seminar on Information and Communication Technology Statistics 19-21 July 2010, Seoul, Korea. Available at https://unstats.un.org/unsd/economic_stat/ICTKorea/Documents/7.1_Balboni_EC_LAC.pdf
- Chikaire, J. U., Godson-Ibeji C. C. & Anyaoha N. O. (2017). Assessing the Effects of E-Wallet Scheme in Farm Inputs Distribution to Rural Farmers in Imo State, Nigeria. Retrieved November 2, 2016 from: http://www.e3journals.org/E3_Journal_of_Agricultural_Research_and_Development. 6(2): 034-041
- Dunne, N. (2016). How Technology will Change the Future. Retrieved from: <https://www.weforum.org/agenda/2016/02/the-future-work/E-agriculture> (2016 June). Information Technology in Agriculture, Retrieved from: http://en.wikipedia.org/wiki/ICT_in_agriculture.
- Gautam, Y. and Andersen, P. (2017). Rural livelihood diversification and household well-being: Insights from Humla, Nepal. *J Rural Stud.*, 44:239–249.

- Igwe, P.A., Amaugo, A.N. and Ogundana, O.M. (2018). Factors affecting the investment climate, SMEs productivity and entrepreneurship in Nigeria. *Eur J Sust Dev.*, 7: 1–19.
- Kassie, G.W. and Ebissa, T.T. (2017). Determinants of Households Saving Behaviour: Case of East Gojjam Zone, Ethiopia. *International Journal of Research in Finance and Marketing*, 7(2): 93-102.
- Lokeswari K. A (2016). Study of the Use of ICT Among Rural Farmers. *International Journal of Communication Research*. 2(6):232- 238.
- Machina, H. and Lubungu, M. (2018). Understanding Intra-household Gender Disparities of Smallholder livestock Veterinary in Zambia. *Journal of Gender, Agriculture and Food Security*, 4 (20):11-24. <https://doi.10.19268/JGAFS.422019.2>.
- Mtega, W.P and Msungu, A.C. (2013). Using Information and Communication Technologies for Enhancing, the Accessibility of Agricultural Information for Improved Agricultural Production in Tanzania. *The Electronic Journal on Information Systems in Developing Countries* 56(1): 1-14.
- Njuguna, M. I., Munyua, N. C. & Makal, K. S. (2015). Influence of Demographic Characteristics on Adoption of Improved Potatoes Varieties by Smallholder Farmers in Mumberes Division, Baringo County, Kenya. *Journal of Agricultural Extension and Rural development*, 7(4): 114-121.
- Nuer, T. K. A. (2018). Why invest in ICTs for agriculture? (CTA Discussion Paper). CTA. Palmer, T. mAgri programme: Orange Sènekèla Midline (GSMARreport).
- Norsida, B. M.,Jasim, M. S., Salim, H., Falah, H. Z., Nolila M N. and Sulaiman, U. (2016). Training Needs of Agricultural Extension Agents Using Borich Needs Assessment Model. *Asian Journal of Agricultural Extension, Economics & Sociology*. 13(1): 1-19.
- NPC, (2006). National Population Commission. Project figures of Bauchi State 2018.
- Nwafor, C.U., Ogundeji A.A., & van der Westhuizen, C. (2020). Adoption of ICT-Based Information Sources and Market Participation Among Smallholder Livestock Farmers in South Africa. *Agriculture*,10(2): 44. (Crossref), Google Scholar.
- Nwokoro, C.V. and Chima, F.O. (2017). Impact of Environmental Degradation on Agricultural Production and Poverty in Rural Nigeria. *Am Int J Contemp Res.*, 7: 6–14.
- Odoemenem, I.U. & C.P.O. Obine (2015). Assessing the Factors Influencing the Utilization and Adoption of Improved Cereals Crop Production Technologies by Small-scale farmers in Nigeria. *India Journal of Science and Technology*. 3(1): 874- 946.
- Olaniyi, O.A. & Adewale J.G. (2014). Women Farmers' Perception and Utilization of Marketing Information on Cassava in Ogun State Nigeria. *Journal of Agricultural Extension*, 18:25-

- Onasanya, S.A., Shehu, R.A., Ogunlade, O.O, and Adefuye, A.L. (2011). Teachers; Awareness and Extent of Communication and Technologies for Effective Science and Health Education in Nigeria. *Singapore Journal of Social Research*. **1**(1):49-58.
- Onemolease, E.A., Ehilenboadiaye, C.O., & Omoregie, O., (2021). Perception of the Role of Agricultural Extension Services by Cassava Processors in Edo State, Nigeria. *ADAN Journal of Agriculture*, **2**(1):70-80.
- Onwutuebe, C. J. (2019). Patriarchy and Women Vulnerability to Adverse Climate Change in Nigeria. <https://doi.org/10.1177/2158244019825914>Sage.
- Pande, N. & Deshmukh, P. (2015). ICT: A Path Towards Rural Empowerment Through Telecommunication, E-governance, and E-Agriculture. *IBMRD's Journal of Management and Research*, **4**(2): 47-54.
- Rehman, F., Muhammad, S., Ashraf, I., Chaudhary, K.M., Ruby, T. & Bibi, I. (2016). Effect of farmers' socioeconomic characteristics on access to agricultural information: Empirical evidence from Pakistan. *The Journal of Animal & Plant Sciences*, **23** (1): 324-329. Retrieved: (7/12/2021) www.cropin.com/ict-in-modern-agriculture: Information and Communication Technologies (ICTs).
- Richardson, D. (2006). The Internet, Rural and Agricultural Development: An Integrated Approach. FAO, Rome <http://www.fao.org/docrep/w6840e/w6840e00.htm>
- Thomas, B.U., Zhuany, J., Owuso, S.A. & Ayamba, E.C. (2017). Factors Affecting the Agricultural Technologies Adoption: The Case of Improved Rice Varieties in Northern Region of Ghana. *Journal of Economics and Sustainable Development*. **8** (8):137-148.
- Titilope, A. O. (2017) Digital Divide and its Socio-Psychological Implications on Rural Dwellers in Nigeria, *International Journal of Innovation in the Digital Economy*. **9**(1):2 6-34. doi:10.4018/ijide.2018010103.
- Yekini, O.T. and Hussein, L.A., (2007). "An Assessment of the Relevance of Information and Communication Technologies (ICTs) to Agriculture and Rural Development by Research and Extension Personnel in South- Western Nigeria". *Nigerian Journal of Rural Sociology*. **8**(1):79-88.
- Zhuang, J. & Naminse, E.Y. (2018). Does Farmer Entrepreneurship Alleviate Rural Poverty in China? Evidence from Guangxi Province. *PLoS ONE*, **13**: 1–18.